Power Plant Energy Prediction using ANN

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Tech: PyTorch | Gradio | ML

Problem Statement

Predict energy output of power plant

Based on environmental conditions

Helps in optimization & efficiency

Dataset & Features

Dataset: Power Plant Data

Features:

- Temperature
- Vacuum
- Pressure
- Humidity

Target: Produced Energy

Methodology

01

Data preprocessing & scaling

02

Tensor conversion

03

TensorDataset + DataLoader

04

ANN model building

Model Architecture

Input Layer (4 features)

Hidden Layers (ReLU activation)

Output Layer (1 neuron)

Loss: MSE

Optimizer: Adam

Results

Achieved ~93% accuracy

Efficient prediction performance

Stable training using batch learning

able training using batch learning

Deployment

- Built Gradio UI
- Real-time prediction
- User-friendly interface
- Ready for production

Conclusion & Future Work

End-to-end ML pipeline completed

Future:

- Model explainability
- Cloud deployment
- Advanced UI enhancements